

Lab Problem Statement 6: Statistical Analysis of Physical Characteristics of Palmer Penguins

Topics: Descriptive Statistics, Hypothesis Testing, ANOVA & Non-Parametric Tests

Problem Statement

Researchers studying penguins in the Palmer Archipelago, Antarctica, collected measurements from different penguin species to understand variations in their physical characteristics.

Using the **Palmer Penguins dataset**, perform a comprehensive statistical analysis in **R** to investigate whether physical characteristics such as **body mass and flipper length differ significantly across penguin species and sex**.

The analysis should combine descriptive statistics, hypothesis testing, effect-size estimation, ANOVA, post-hoc analysis, and appropriate non-parametric tests.

Dataset

Dataset

[UCI Machine Learning Repository – Palmer Penguins Dataset](#)

The dataset contains observations of **344 penguins** belonging to three species:

- Adélie
- Chinstrap
- Gentoo

Important variables include:

- species – Penguin species
- island – Island where the penguin was observed
- bill_length_mm – Bill length in millimetres
- bill_depth_mm – Bill depth in millimetres
- flipper_length_mm – Flipper length in millimetres
- body_mass_g – Body mass in grams
- sex – Sex of the penguin

Tasks

1. Descriptive Statistical Analysis

Perform descriptive analysis of **body mass (body_mass_g)**.

Calculate:

- Mean
- Median
- Minimum and maximum

- Variance
- Standard deviation
- First and third quartiles
- Interquartile range (IQR)
- Skewness
- Kurtosis

Compute appropriate descriptive statistics separately for different penguin species.

Visualize the data using:

- Histogram
- Boxplot
- Density plot

2. Hypothesis Testing

Investigate whether **male and female penguins have significantly different mean body masses**.

1. State the null and alternative hypotheses.
2. Check the normality assumption using:
 - Shapiro-Wilk test
 - QQ-plots
3. Perform an appropriate **independent two-sample t-test**.
4. Calculate a **95% confidence interval** for the difference between mean body masses.
5. Calculate **Cohen's d** to measure the effect size.
6. Interpret the p-value, confidence interval, and effect size.

3. One-Way ANOVA

Investigate whether **mean body mass differs among the three penguin species**.

Before performing ANOVA:

- Examine the distribution of body mass within each species.
- Check normality using QQ-plots and/or the Shapiro-Wilk test.
- Check homogeneity of variance using **Levene's test**.

Perform:

One-way ANOVA:

`body_mass_g ~ species`

Report and interpret:

- F-statistic

- Degrees of freedom
- p-value

If the ANOVA result is statistically significant, perform **Tukey's HSD post-hoc test** to identify which pairs of penguin species differ significantly.

4. Non-Parametric Analysis

If the assumptions of one-way ANOVA are substantially violated, perform the **Kruskal-Wallis test** using:

```
body_mass_g ~ species
```

Compare the results obtained using:

- One-way ANOVA
- Kruskal-Wallis test

Discuss whether both methods lead to similar conclusions.

5. Two-Way ANOVA

Investigate the simultaneous effect of **species and sex** on penguin body mass.

Perform a two-way ANOVA using:

```
body_mass_g ~ species * sex
```

Determine whether:

- Penguin **species** significantly affects body mass.
- Penguin **sex** significantly affects body mass.
- There is a significant **interaction between species and sex**.

Interpret the results in the context of penguin physical characteristics.

6. Additional Analysis

Repeat an appropriate part of the analysis using **flipper length (flipper_length_mm)** instead of body mass.

Investigate whether flipper length varies significantly among the three penguin species and discuss whether the findings are consistent with those obtained from body mass.

7. Visualization and Interpretation

Prepare suitable graphical representations, including:

- Histogram of body mass
- Species-wise boxplot of body mass
- Sex-wise boxplot of body mass
- QQ-plots for checking normality
- Species-wise flipper-length comparison
- Group comparison plot showing significant differences among species

Provide appropriate titles, axis labels, legends, and interpretations for all graphs.

Deliverables

Students must submit:

1. **R Script (.R)** containing the complete analysis.
 2. Descriptive statistical summary.
 3. Hypothesis-testing results with appropriate interpretation.
 4. ANOVA table containing F-statistic and p-value.
 5. Tukey HSD results, wherever applicable.
 6. Kruskal-Wallis test results.
 7. Appropriate statistical visualizations.
 8. A brief **R Markdown/PDF/HTML report** explaining the statistical findings, p-values, confidence intervals, effect sizes, and conclusions.
- SUBMIT COLAB NOTEBOOK PDF
 - ALSO UPLOAD THE SUBMISSION ON GIT ACCOUNT AND SHARE LINK OF GIT UPLOAD IN SUBMISSION

I hereby declare that:

1. The assignment / examination has been designed in alignment with the approved curriculum and intended Learning / Course Outcomes.
2. The assessment incorporates the following key elements:
 - o Problem-solving ability
 - o Critical reasoning
3. An AI tool has been used only for reviewing and enhancing assessment quality (where applicable) and not as a replacement for academic judgment.
4. The assignment has been reviewed, finalized, and approved by me, and I take full academic responsibility for its content, rigor, fairness, and integrity.
5. I understand that submission of this declaration is mandatory for all assignments and that it must be presented during the Meeting with the Chief Quality Officer (CQO) or any quality audit.

AI Feedback Scores

Rating: 1 – Lowest | 4 – Highest

Parameter	Score
Real-life Application	4 / 4
Problem-solving Ability	4 / 4

Parameter	Score
Critical Reasoning	4 / 4
Creativity	3 / 4
Alignment with Learning / Course Outcomes	4 / 4

Estimated Time Required

1 Hour 45 Minutes – 2 Hours

Lab 1: 50–55 Minutes

- Compute descriptive statistical measures
- Visualize distributions using boxplots
- Perform paired and two-sample t-tests
- Calculate confidence intervals and effect size
- Interpret statistical significance and summarize findings

Lab 2: 55–65 Minutes

- Verify ANOVA assumptions using statistical tests
- Perform One-way and Two-way ANOVA
- Conduct Tukey HSD and Kruskal-Wallis tests where appropriate
- Generate diagnostic plots and comparison visualizations
- Interpret statistical outputs and prepare analytical reports

COs Linked with the Assignment

CO2, CO3, CO6

LOs Linked with the Assignment

LO 3.1, LO 3.2, LO 3.3, LO 4.1, LO 4.2, LO 6.1

Justification

This laboratory assignment provides students with practical experience in **statistical data analysis and hypothesis testing** using realistic pharmaceutical and organizational analytics case studies. The problems are designed around authentic experimental datasets that require students to perform descriptive statistics, inferential statistical analysis, validate statistical assumptions, and interpret analytical outcomes for evidence-based decision-making. These activities closely mirror the statistical analysis workflows used in biomedical research, clinical trials, and organizational performance evaluation.

The first laboratory introduces students to descriptive statistics and hypothesis testing by analyzing the effectiveness of a cholesterol-lowering drug. Students compute measures of central tendency and dispersion, examine distribution characteristics using skewness and kurtosis, perform paired and independent sample t-tests, calculate confidence intervals and effect sizes, and interpret statistical significance. These tasks strengthen students' understanding of inferential statistics while emphasizing correct interpretation of experimental results.

The second laboratory extends statistical reasoning by requiring students to evaluate multiple experimental groups using **One-way ANOVA**, **Two-way ANOVA**, **Tukey HSD**, and **Kruskal-Wallis** tests. Students first verify statistical assumptions through normality and homogeneity tests before selecting the appropriate analytical technique, thereby reinforcing the importance of evidence-based statistical decision-making. The assignment also promotes visualization and interpretation of statistical findings through diagnostic plots and comparative graphs.

The assignment demonstrates strong alignment with **CO2** by enabling students to apply statistical concepts, hypothesis testing techniques, and analytical visualization using R. It significantly supports **CO3** through practical implementation of statistical analysis, experimental evaluation, and interpretation of real-world datasets. Furthermore, the complete workflow—from data exploration and statistical testing to interpretation, visualization, and report preparation—partially contributes to **CO6** by introducing students to structured end-to-end statistical analysis projects. The inclusion of real-life healthcare and workplace analytics scenarios, assumption validation, effect size estimation, and professional reporting practices enhances students' analytical reasoning, statistical literacy, and problem-solving skills while preparing them for advanced data science and research-oriented applications.

AI Tool Used

ChatGPT

Prompt(s) Used

"Evaluate the given R Programming laboratory assignment using a 1–4 rubric for Real-life Application, Problem-solving Ability, Critical Reasoning, Creativity, and Alignment with Learning / Course Outcomes. Identify the estimated time required, map the relevant Course Outcomes (CO1–CO6) and Learning Outcomes (LOs), and justify the ratings based on the R Programming syllabus covering descriptive statistics, hypothesis testing, confidence intervals, effect size, ANOVA, non-parametric tests, statistical visualization, and interpretation of experimental data."